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CAD/CAM system design

Parametric Model of a Hexagonal Nut

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1 Introduction

The subject of this project is a parametric model of a hexagonal nut. The description is based on **ISO 4032:2012** [2], **ISO 225:2010** [1] and **ISO 68-1:1998** [3].

The model was created in FreeCAD with all dimensions being driven by a spreadsheet, enabling the nut to be easily reconfigured.

2 Reference Standards

- **ISO 4032:2012** — Hexagon regular nuts (style 1) – Product grades A and B
- **ISO 225:2010** — Fasteners – Bolts, screws, studs and nuts – Symbols and descriptions of dimensions
- **ISO 68-1:1998** — ISO general purpose screw threads — Basic profile — Part 1. Metric screw threads

3 Parameters

The primary input is the metric thread designation (e.g. M10). Selecting a thread type populates a lookup table derived from **ISO 4032:2012**. Values for M10 thread are presented in table 1. After the thread type is selected, any individual parameter can be overridden. Thread profile dimensions computed from the pitch P are presented in table 2.

Angular parameters are set to values presented in table 3.

Table 1: Parameters and their default values for M10 thread type (dimensions in millimetres; nominal/max)

Parameter	Symbol	M10
Basic major diameter (nominal diameter) of thread	D	10
Pitch	P	1.5
Diameter of the countersink	d_a	10.80
Outer diameter of the bearing face	d_w	14.60
Width across corners	e	17.77
Nut height	m	8.40
Width across flats	s	16.00

Table 2: Thread profile dimensions derived from pitch P (dimensions in millimetres)

Parameter	Formula	Value for M10 ($P = 1.5$)
Fundamental triangle height	$H = \frac{\sqrt{3}}{2} P$	1.299
Thread depth (nut)	$\frac{5}{8} H$	0.812
Crest truncation	$\frac{H}{8}$	0.162
Root truncation	$\frac{H}{4}$	0.325

Table 3: Angular parameters.

Symbol	Description	Value
θ	Countersink angle	120°
β	External chamfer angle	30°
Thread flank angle	–	60°

4 Technical Drawing

Figure 1 presents an annotated technical drawing of the hexagonal nut with the key dimensions labelled.

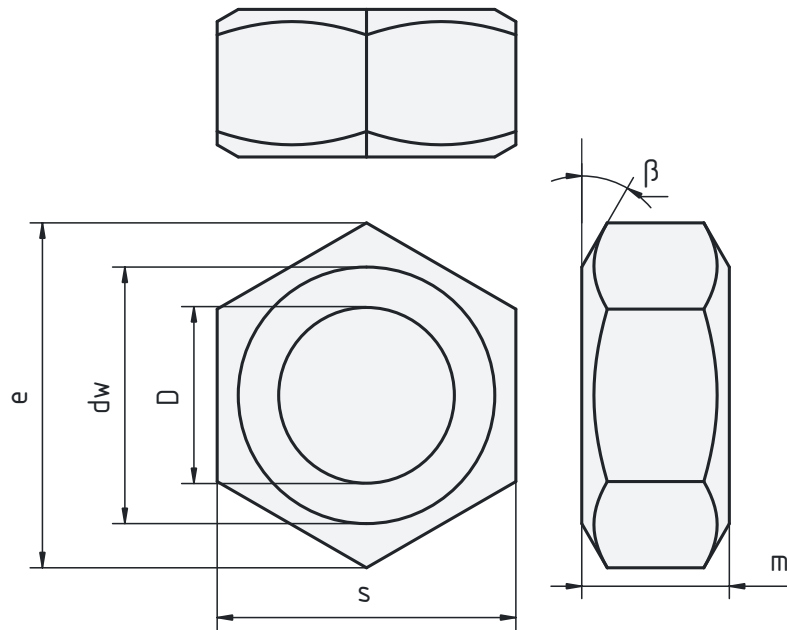


Figure 1: Technical drawing of the hexagonal nut with parameter annotations

References

- [1] *Fasteners — Bolts, screws, studs and nuts — Symbols and designations of dimensions*. ISO, 2010.
- [2] *Hexagon regular nuts (style 1) — Product grades A and B*. EN ISO, 2012.
- [3] *ISO general purpose screw threads — Basic profile — Part 1*. ISO, 1998.